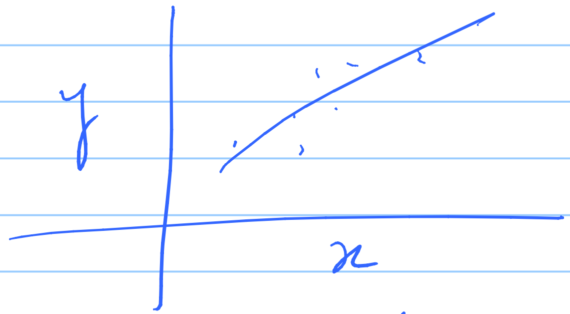
Note Title 8/20/2025

COL774 / COL7341
Machine Learning
Aug 20, 2025

Last Class: Newton's Method
Closed form solution for Least Squares Regression



$h_\theta(x) = \theta^T x$

$$J(\theta) = \underbrace{\frac{1}{2m}\sum_{i=1}^{m}\left[y^{(i)} - \theta^T x^{(i)}\right]^2}$$

convex fn. as a fn of θ

$\arg\min_{\theta} J(\theta)$

↳ $\nabla_\theta J(\theta)$ ⤳ use gradient descent

Gradient Update ← $\{\ \theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \cdot \nabla_\theta J(\theta)$

+ve semi-definite: $A \in \mathbb{R}^{m\times n}$ is symmetric

$z^T A z \geq 0$

Question:- Can we directly find θ s.t $J(\theta)$ is minimized?

or equivalently θ:- $\nabla_\theta J(\theta) = 0$

$\{x^{(i)}, y^{(i)}\}_{i=1}^{m}$

$X \in \mathbb{R}^{m \times n} = \begin{bmatrix} \text{---} & x^{(i)T} & \text{---} \end{bmatrix}$ $\quad y = \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(m)} \end{bmatrix} \quad \theta = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_n \end{bmatrix}$

Design Matrix

$x^{(i)} \in \mathbb{R}^n$

$y - X\theta \quad \cancel{(X\theta - y)} = \begin{bmatrix} | \\ (y^{(i)} - \theta^T x^{(i)}) \\ | \end{bmatrix}$

$(y - X\theta)^T (y - X\theta) = \sum_{i=1}^{m} \left[y^{(i)} - \theta^T x^{(i)}\right]^2 = 2mJ(\theta)$

$J(\theta) = \frac{1}{2m} (y - X\theta)^T (y - X\theta) = \frac{1}{2m} \Big[\overset{1}{y^T y} - \overset{2}{(X\theta)^T y} - \overset{3}{y^T X\theta} + \overset{4}{(X\theta)^T X\theta} \Big]$

$$y^T X \theta = (y^T X \theta)^T$$
$$= \theta^T (y^T X)^T$$
$$= \theta^T X^T y$$

$$= \frac{1}{2m}\left[y^T y - \underbrace{\theta^T X^T y}_{2} - \underbrace{\theta^T (X^T y)}_{3} + \theta^T X^T X \theta \right]$$

$$= \frac{1}{2m}\left(\underset{1}{y^T y} - \underset{2}{2\theta^T (X^T y)} + \underset{3}{\theta^T (X^T X)\theta} \right]$$

$$\Rightarrow \nabla_\theta J(\theta) = \frac{1}{2m}\left[\underbrace{\nabla_\theta (y^T y)}_{0} - \underbrace{\nabla_\theta\, 2\theta^T (X^T y)}_{2X^T y} + \underbrace{\nabla_\theta \left[\theta^T (X^T X)\theta\right]}_{2(X^T X)\theta} \right]$$

$$\Rightarrow \nabla_\theta J(\theta) = \frac{1}{2m}\left[-2X^T y + 2(X^T X)\theta \right]$$

Equate it to zero:$\Rightarrow$

$$\Rightarrow \frac{1}{2m}\left(-2X^T y + 2X^T X\theta \right] = 0$$

$$\Rightarrow \cancel{2}(X^T X)\theta = \cancel{2} X^T y$$

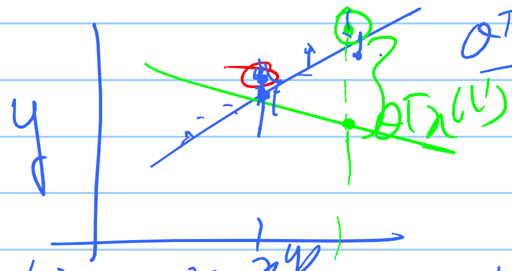
$$\Rightarrow \quad \theta = \underbrace{\left[(X^T X)^{-1} X^T\right]}_{\text{Pseudo inverse}} y$$

Probabilistic Interpretation:- of Least Squares:-] Maximum Likelihood Estimators (MLE's)

Generative Modeling of Data

$f : X \rightarrow Y$

$\sim h_\theta$



$\theta^T x = h_\theta(x)$

$\{x^{(i)}, y^{(i)}\}_{i=1}^{m}$

$$\theta^T x^{(i)} + \varepsilon_\theta^{(i)} = y^{(i)} \qquad \varepsilon_\theta^{(i)} \sim N(0, \sigma^2)$$

$\varepsilon_\theta^{(i)}$ i.i.d.

$$y^{(i)} \mid x^{(i)}; \theta \longrightarrow y^{(1)}, \ldots, y^{(m)}$$

density $\swarrow$ $\underline{P(y^{(i)}|x^{(i)};\theta)}$ $= ?$ $\qquad N(\theta^T x^{(i)}, \sigma^2)$

$$= \frac{1}{(2\pi\sigma^2)^{1/2}} e^{-\frac{[y^{(i)} - \theta^T x^{(i)}]^2}{2\sigma^2}} \quad — (1)$$

Likelihood of data $=$
$L(\theta)$

$$\underbrace{P(y^{(1)} \ldots y^{(m)} \mid x^{(1)} \ldots x^{(m)}; \theta)}$$

$$= \prod_{i=1}^{m} P(y^{(i)}|x^{(i)};\theta) \quad — (2)$$

given $(x^{(i)}, y^{(i)})$ are i.i.d's

$$\log \prod_{i=1}^{m} P(y^{(i)}|x^{(i)},\theta) = \underbrace{\sum_{i=1}^{m} \log P(y^{(i)}|x^{(i)};\theta)} = LL(\theta)$$

$\Rightarrow$ $\underline{MLE}$ is $\operatorname{argmax}_{\theta} LL(\theta)$ $\qquad$ log-likelihood of data

$$\sum_{i=1}^{m} \log \left[\frac{1}{(2\pi\sigma^2)^{1/2}} \, e^{-\frac{[y^{(i)} - \theta^T x^{(i)}]^2}{2\sigma^2}} \right]$$

$$= \underbrace{\sum_{i=1}^{m} \log \frac{1}{(2\pi\sigma^2)^{1/2}}}_{\text{Constant}} + \underbrace{\sum_{i=1}^{m} - \frac{[y^{(i)} - \theta^T x^{(i)}]^2}{2\sigma^2}}_{-\frac{J(\theta)(\cancel{2}m)}{\cancel{2}\sigma^2}}$$

$$= C - \frac{J(\theta)\, m}{\sigma^2}$$

$$\underset{\theta}{\arg\max}\; LL(\theta) \equiv \underset{\theta}{\arg\max} \; -\frac{J(\theta)\, m}{\sigma^2} \equiv \underline{\underset{\theta}{\arg\min}\; J(\theta)}$$